

GRANT FELLOWS

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EDUCATION

Bachelor of Science in Mechanical Engineering, Purdue University

Expected May 2027

SKILLS

3D Modeling & Simulation	Siemens NX, Inventor, Fusion 360, Simcenter/FEMAP (CAD/CAM/FEA)
Fabrication	Manual mill/lathe; CNC mill (Tormach), 3D printing, laser cutting
Software Development	Python, Java, JavaScript, MATLAB; Flask, HTML, CSS; ROS 2

EXPERIENCE

Undergraduate Research Assistant, Applied Optics Lab, Purdue University May 2026 - Present
West Lafayette, Indiana

- Designing, fabricating, and testing a 3-axis stepper-driven gantry robot with servo gripper for automated biological sample handling in Prof. Euiwon Bae's group; built Raspberry Pi/Python control stack for motors and end effector

Software and Autonomy Lead, Purdue Vertical Flight Systems Aug 2025 - Present
West Lafayette, Indiana

- Developing ROS 2 nodes for flight control, sensor integration, and obstacle avoidance on an autonomous emergency response aircraft (GoAERO Prize); built obstacle detection pipeline from stereo camera point clouds
- Conducted flight test operations, planning and executing tests to validate autonomy software on hardware

OrbShip Fellow, Orb Aerospace May 2025 - Aug 2025
Grand Rapids, Michigan

- Developed a Python library and web app improving FEM simulation of additively manufactured parts, letting Orb engineers predict stiffness and failure of printed parts before physical testing; presented approach to Orb team
- Conducted 50+ ASTM tensile tests and data analysis to demonstrate validity of simulation in real-world parts

Systems Engineering Lead, Purdue Lunabotics Aug 2024 - May 2025
West Lafayette, Indiana

- Led systems engineering for Purdue's entry into the NASA Lunabotics competition, tracking and verifying system requirements; placed 5th in Systems Engineering, and 3rd at the UCF Qualifier (2025)
- Previously a systems team member (2023–24) on a team awarded 3rd in Systems Engineering, 2nd in Presentation & Demonstration, and 4th overall out of 58 teams

Laser-Assisted Processing Researcher, Purdue University Aug 2023 - May 2024
West Lafayette, Indiana

- Synthesized 100+ papers on laser-assisted processing for industrial decarbonization into a report and final presentation guiding further research by Dr. Benxin Wu's Vertically Integrated Projects team

PUBLICATIONS

X. You, V. Cattaneo, S. Yanagisawa, S. B. Larson, M. C. Lee, R. R. Kurup, **G. Fellows**, A. K. Swadi, A. R. Chandra, and T. A. Guimarães, "Development of a High-Payload Electric Vertical Takeoff and Landing Multirotor for the GoAERO Competition," *AIAA AVIATION Forum*, 2026, doi:10.2514/6.2026-4585

PROJECTS

Selected personal projects (full portfolio at gwfellows.github.io/portfolio):

- A GPS mount for ship railings. Built for a container ship captain's custom use case; parts are in active use today
- A Python library for tree-based genetic programming, used to perform symbolic regression on datasets